9. Protocol T=0, asynchronous half-duplex character transmission
This agreement starts after "Answer to Reset" or "Successful PPS Exchange".

9.1 Command header
The interface device tells the card what command to start by sending a 5-byte header. Called the command header. The command header is a series of 5 bytes, these 5 bytes are designated as CLA, INS, PI, P2, P3.
• CLA: instruction category, "A0" represents the command category, and "FF" represents the header of the PPS process;
• INS: instruction code, the instruction code is only valid when the bits b8-b5 are not equal to ‘6’ and ‘9’;
• P1, P2: instruction additional parameters (such as address);
• P3: Determined by the code of the INS, or indicates the length of the data sent to the SIM card in the command, or the maximum length of the data waiting for a response from the SIM card.
---- In the transmission command of output data, P3=0 means that 256 bytes are transmitted from the card.
---- In the transmission command of input data, P3=0 means not to transmit data from the card.
Example: A0A40000023F00: Enter the 3F00 file, where A4 is the command, which is equivalent to using cd to enter the file position in the terminal.
File identifier: 3F: main file; 7F: first level proprietary file; 5F: second level proprietary file;
• 2F: basic files under the main file; 6F: basic files under the first level of proprietary files;
• 4F: Basic documents under the second level of proprietary documents.

9.2 Process byte
The value of the process byte will indicate the action requested by the interface device, that is, the response of the smart card. There are three types of process bytes.
In each process byte, the smart card can use an ACK or NULL byte to continue the command, or express disapproval with an appropriate non-response, or end the command with the end sequence SW1-SW2.

字节	值	有效的数据传输	接受
NULL	60	无	一个过程字节
ACK	INS	所有剩余的数据字节	一个过程字节
	INS⊕FF	下一个数据字节	一个过程字节
SW1	6X，9X（不包含 60）	无	一个过程字节

• NULL is equal to '60': NULL request does not affect the data transmission, only waiting for one process byte.
• ACK. In ACK, bit b8 b2 should be the same as b8 in INS b2 is the same or complementary, but the value of these bits of INS should not be equal to '6x' and '9x'.
① If ACK=INS, all remaining bytes will be transmitted subsequently;
② If the exclusive OR of ACK and INC=0xFF, then only the next byte will be transmitted.
• SW1 is equal to ‘6x’ or ‘9x’ but not equal to ‘60’.
After receiving SW1, the interface device will wait for a SW2 byte transmission. There is no restriction on the value of SW2.

The end command SW1-SW2 gives the card status at the end of the command. Setting it to 9000 just completes the process normally.
When the upper 4 bits of the effective bits of SW1 are equal to 6, the meaning of SW1 has nothing to do with the application.
The difference between T=0 and 1: The most obvious difference between the protocols is that the T=1 protocol is layered according to the OSI reference model, which are: physical layer, data link layer, and application layer. The physical layer is mainly the transmission of data characters, the data link layer is mainly the transmission of data blocks, and the application layer is mainly the interactive transmission of APDUs. The interaction of APDU and the transmission of data characters are roughly the same as those described in the T=0 protocol. The key is the data block transmission defined by the data link layer. Because of the definition of the data link layer, the T=1 protocol basically has all the features that can realize complex network data transmission. In contrast, the T=0 protocol is simply equivalent to "naked transmission."

9.3 Examples

9.3.1 Determine whether the SIM card is a white card

```
A0A40000023F00 // 进入3F00主文件
A0A40000027F20 // 进入7F20文件
A0A40000026F07 // 进入6F07文件
A0B0000009     // B0 允许SIM卡从当前透明文件中读取字节串
// 返回的字符串
str = [str substringWithRange:NSMakeRange(0, 18)];
    if ([str isEqualToString:@"FFFFFFFFFFFFFFFFFF"]) {
        // 为白卡
        return YES;
    }
```

9.3.2 Read SMS Center

```
A0A40000023F00
A0A40000027F10
A0A40000026F42
A0C000000F
A0B2010428     // B2  用于读取线性固定文件或循环文件的记录
```

9.3.3 Read iccid

```
A0A40000023F00
A0A40000022FE2
A0B000000A
// 返回的卡号为倒序卡号
```

9.3.4 Read IMSI number

```
A0A40000023F00
A0A40000027F20
A0A40000026F07
A0B0000009
```

10 T=0 Transmission example

Code	**Value**
CLA	‘00’
INS	‘20’（Send bytes）
P1	‘00’
P2	‘00’
P3 (Le/Lc)	‘Y’ (y bytes of data)
DATA	Data sent to the card (y Byte)
Le	no
SW1- SW2	‘9000’: normal end

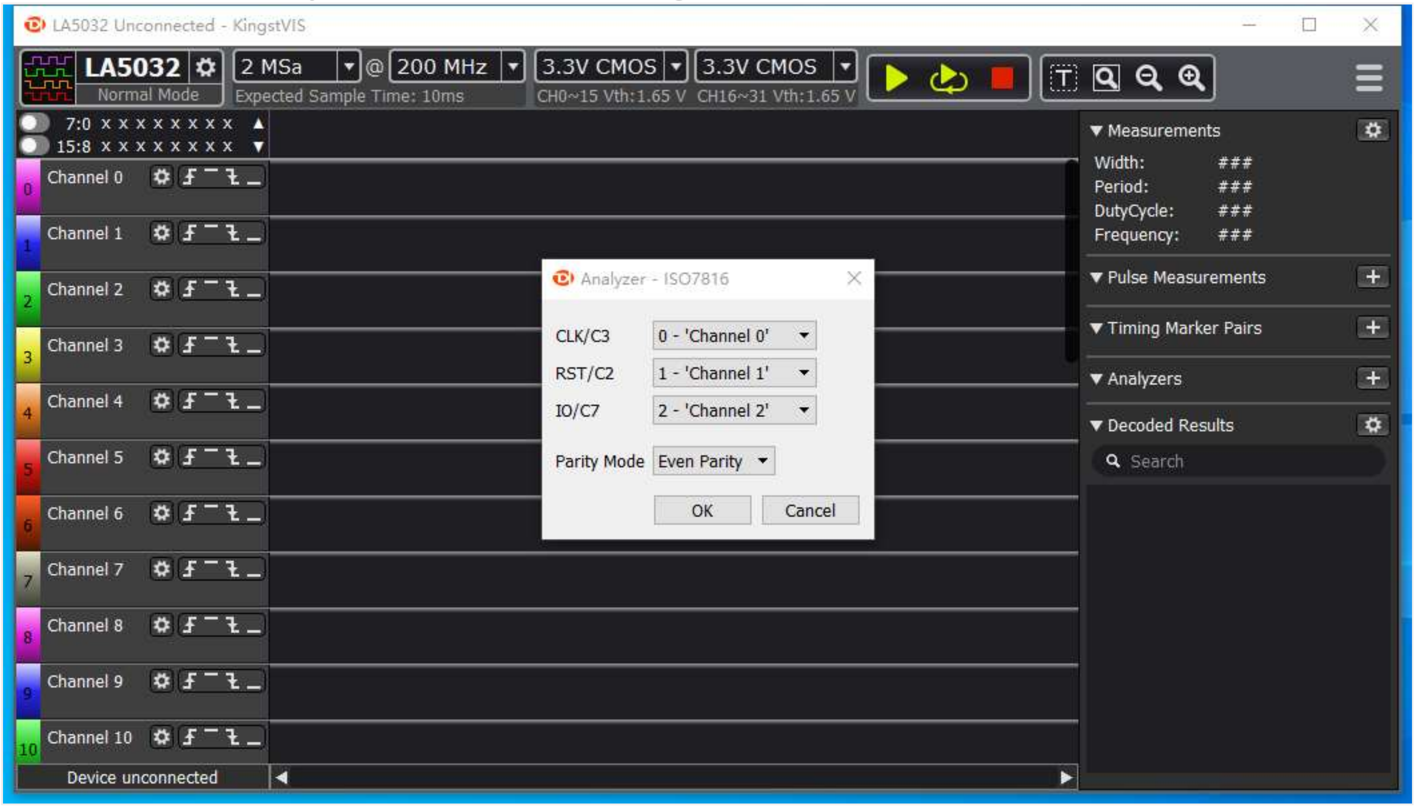
In this example, the card reader will send Lc (y) bytes to the card, and the card will accept these data (the INS value tells cos to execute the received command, and P3 (Lc) is passed as a parameter to the executed function , The function knows how many bytes to receive), and then returns the status (SW). In this APDU, there is no Le, so the card will not send data to the reader (also controlled by cos).

11 T=0 Communication example

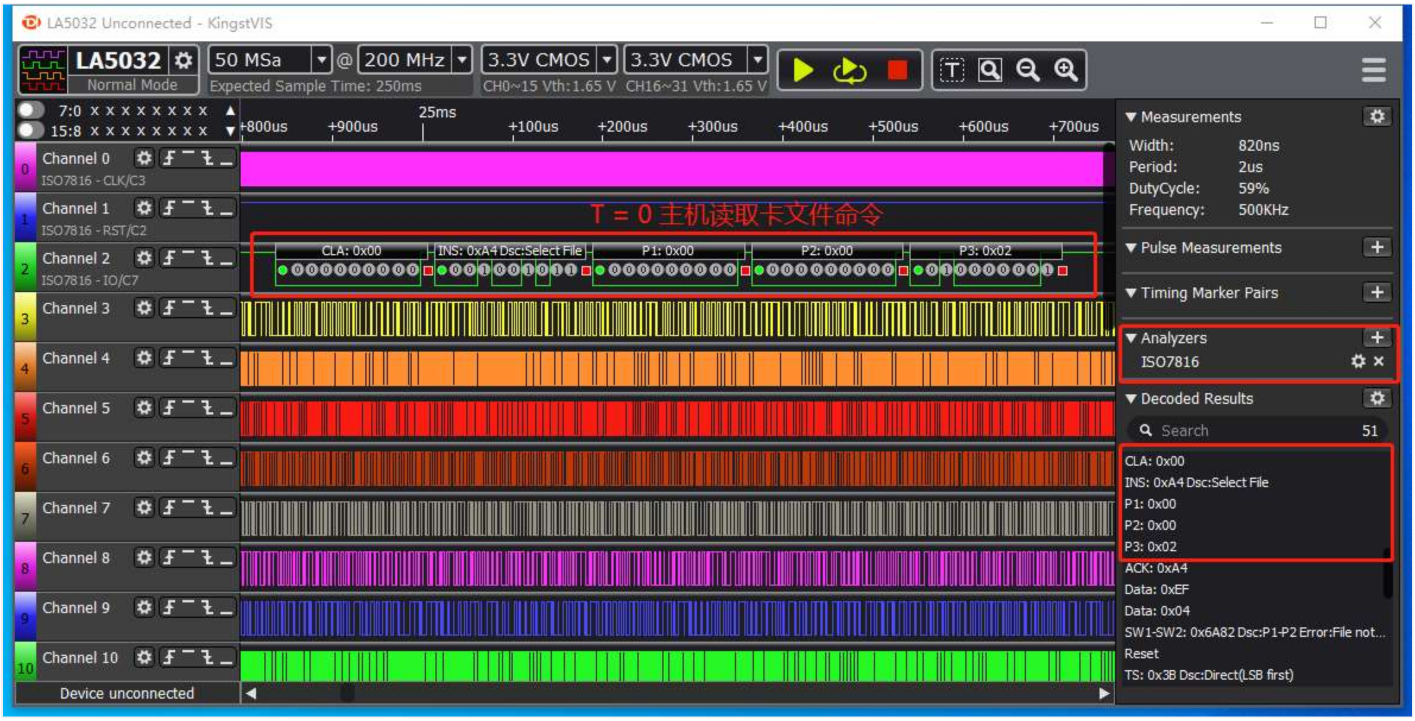
This example uses Jinsite ElectronicsKingst LA5016 usb logic analyzerCheck iso7816 data communication.

The following pictures are screenshots of a complete packet analysis.

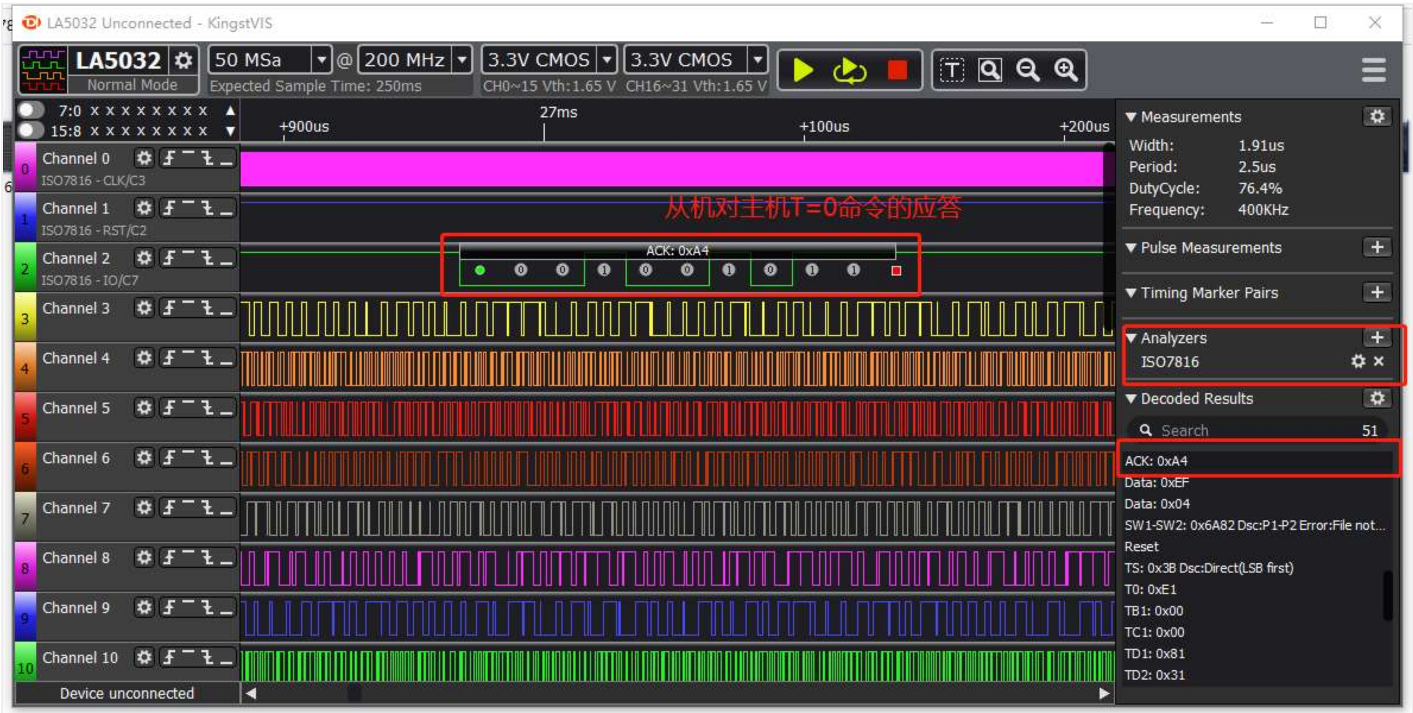
11.1 Protocol analysis parameter setting



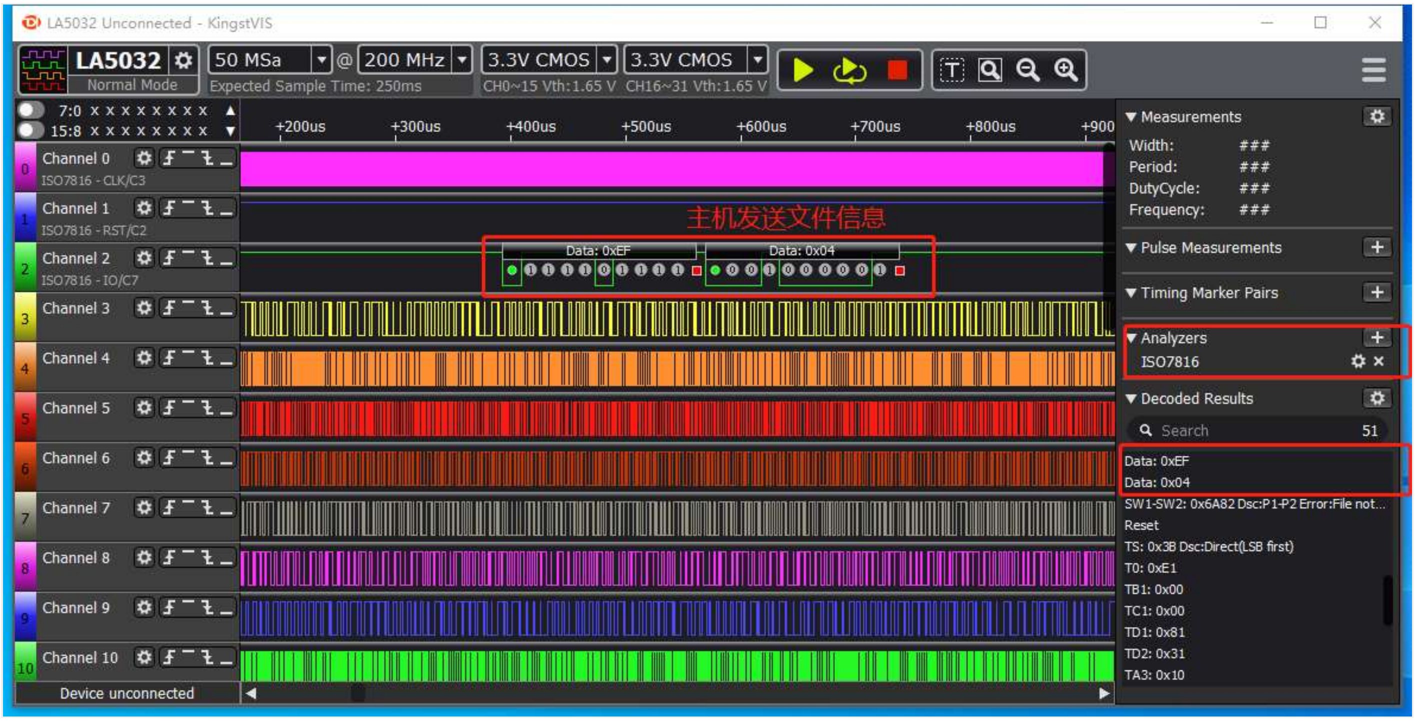
11.2 T=0: According to the INS data, send the command that the host needs to execute, and by parsing P3, the length of the next frame protocol can be obtained



11.3 After receiving the T=0 protocol of the master, the slave returns a response command



11.4 The host sends the file information to be selected



11.5 The slave returns the command execution result